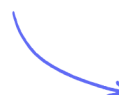


Structured Questions

Atomic Structure

Sub-Atomic Particles / Isotopes & Mass Spectra

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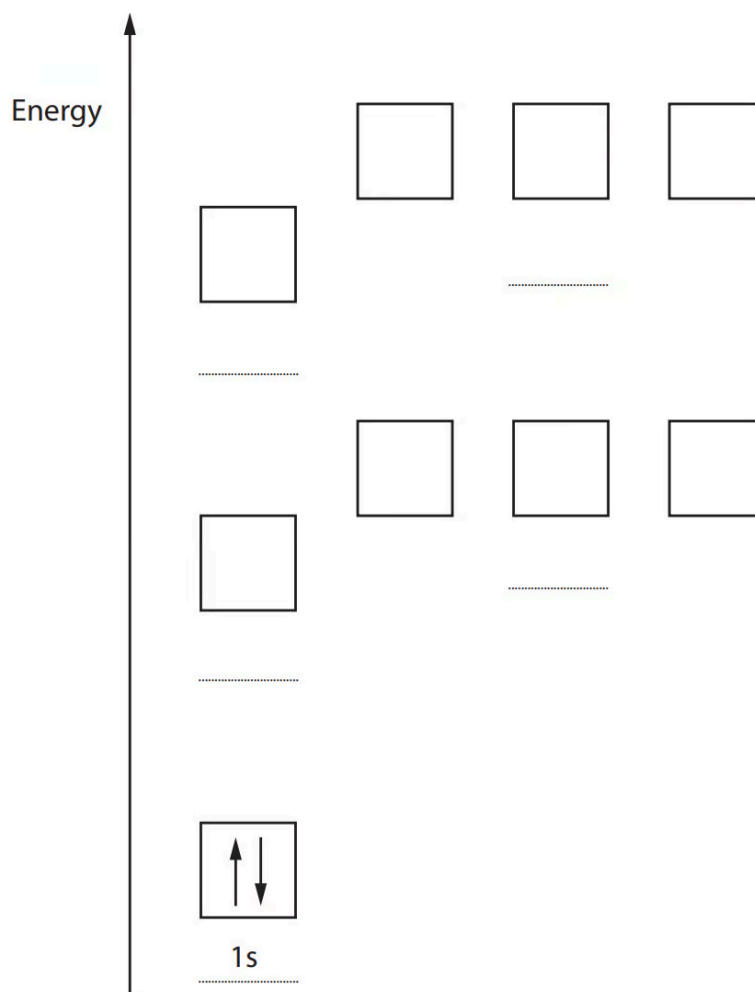


Total Marks

/14

1 (a) This question is about the element sulfur.

Complete the diagram to show the electronic configuration for a sulfur atom in the ground state. Include labels for each subshell.



(2 marks)

(b) Write an equation for the **first** ionisation energy of sulfur. Include state symbols.

(2 marks)

- (c) Explain why the first ionisation energy of sulfur is less than the first ionisation energies of **both** phosphorus and chlorine.

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(3 marks)

- (d) A sample of sulfur contains four isotopes.

Isotope	^{32}S	^{33}S	^{34}S	^{36}S
Percentage abundance	94.88	0.83	4.27	0.02

- i) State what is meant by the term **isotopes**, in terms of subatomic particles.

(2)

- ii) Calculate the relative atomic mass of sulfur in this sample.
Give your answer to **two** decimal places.

(2)

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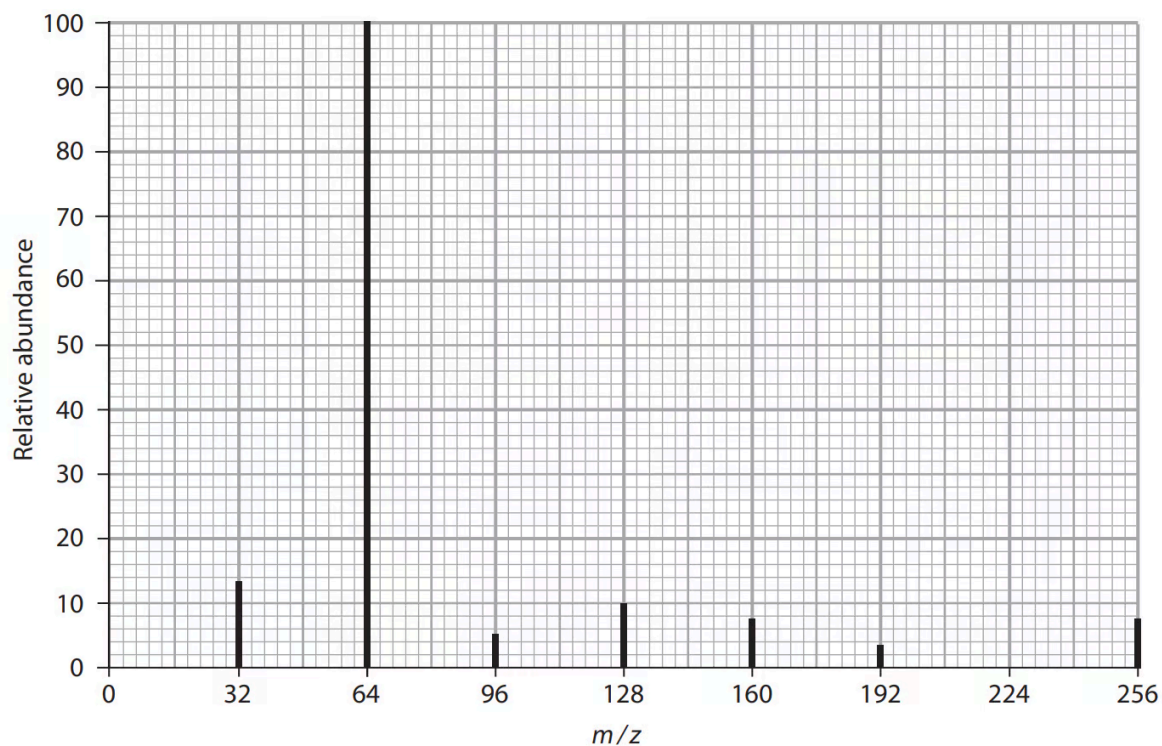
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(4 marks)

(e) The mass spectrum of a sample of sulfur with ^{32}S as the only isotope is shown.



i) Calculate the number of sulfur atoms in the molecular ion.

You **must** show your working.

(1)

ii) Suggest the **formula** of the **most stable ion** shown by this spectrum.

(2)

(3 marks)